

IC 252: Data Science 2

Course Code	IC 252
Course Name	Data Science 2
L-T-P-C	3-0-2-4
Prerequisites	IC 110, IC 152
Intended for	B.Tech 1st
Distribution	Core
Approval	29th BoA (Mar 2019); 20th Senate (Apr 2019)
Revised	47th BoA (Sep 2022) and 37th Senate (Oct 2022) - renamed from "Data Science II"

Course Contents

- **Probability:** Why probability and what is it? (give real life situations which demands use of probability). Counting, combinations, permutations, binomial and multinomial coefficients, Stirling's formula. Discrete probability spaces (with examples). Axiomatic definition of probability, inclusion-exclusion formula, independence, condition probability, Bayes' rule. (Note: Cover the paradoxes and well known problems). [6 Lectures]
Lab: counting, basic probability – simulation of simple experiments, birthday paradox, conditional probability.
- **Random variables:** definition, distribution function and its properties, probability mass function (binomial, Bernoulli, Poisson, geometric), probability density function (uniform, exponential, Gaussian). Joint distributions, independence and conditioning of random variables. Function of random variables, change of variable formula. [9 Lectures]
Lab: Generating random variables following a given pdf/pmf. engineering application of functions of random variables.
- Measures of central tendency, dispersion and association – expectation, median, variance, standard deviation, mean absolute deviation, covariance, correlation and entropy (definition and guidelines on how to choose a particular measure). Markov and Chebyshev inequalities. Notion of convergence in probability and distribution. Weak law of large numbers and central limit theorem (examples demonstrating the use of WLLN and CLT). Montecarlo methods (estimating value of e , π , simulation of birthday paradox). Poisson limit for rare events. [11 Lectures]
Lab: Scatter plot (for independent, correlated, uncorrelated random variables), Montecarlo simulation, WLLN and CLT.
- **Statistics:** Using probability to understand data (give real life examples). Frequentist approach - point and range estimates, confidence intervals, hypothesis testing p-values, significance level, power and t-test. Bayesian inference – maximum likelihood estimation. Regression. [14 Lectures]
Lab: Parameter estimation, hypothesis testing, regression.
- **Case study:** Analyze a large data set (medicine/engineering/biological) using the methods covered in the course. [2 Lectures]

Text Books

1. Sheldon Ross, **Introduction to Probability and Statistics for Engineers**, 5th edition, Elsevier, 2014

References

1. Morris H. DeGroot and Mark J. Schervish, **Probability and Statistics**, 4th edition, Addison-Wesley, 2012
2. Blitzstein and Hwang, **Introduction to Probability**, CRC Press, 2015.
3. William Feller, **An Introduction to Probability**, Volume 1, 3rd edition, Wiley, 2008.
4. Freedman, Pisani, Purves, **Statistics**, 4th edition, W. W. Norton & Company, 2014.